

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

University Examination

Date: 07/05/2018

Day: Monday

Time: 10:00 AM to 10:30 AM

Total Marks: 20

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following
Examination : BPT- May 2018
Semester: 8 th
Section : I
Course Code : PT403.01
Course Name: Physiotherapy in Sports & Allied Therapeutics

Important Instructions:

- There are two sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
- **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
- Marks on the right side suggest the total marks of that question.
- Maximum time allotted for **SECTION – I** is 30 minutes Question paper for **SECTION – II** will be given once the **SECTION – I** is completed within the stipulated time and returned. However, Candidates who complete **SECTION – I** earlier than 30 minutes can collect main question paper for **SECTION –II** immediately after handing over the **SECTION-I** question paper with answers.
- **SECTION – I** should be attached with the main answer book at the end of the examination and should be submitted to the supervisor.

SECTION-I

Multiple Choice Questions: (MCQs) (Encircle the correct answer)

(10 × 1 = 10)

1. The cool down after exercise is important because it:
 - a. Improves oxygen uptake
 - b. Regulates breathing
 - c. Speeds the removal of lactic acid
 - d. Decreases adrenaline
2. Which one of the following is a health screening test?
 - a. Sit and reach test
 - b. 30m sprint test
 - c. Cooper's 12 minute run test
 - d. Romberg's test
3. Which one of the following best indicates a balanced diet
 - a. A balance between what is eaten and energy expended
 - b. A balance of different types of protein
 - c. A balance between what is eaten and what is drunk
 - d. A balance between fruit, vegetables and fiber

4. Which one of the following is an example of muscular endurance?
 - a. To be able to stretch effectively in a gymnastics activity
 - b. To be able to lift a heavy weight correctly in training
 - c. To be able to finish a circuit training session
 - d. To be able to run quickly over a short distance
5. The term energy currency in exercise physiology refers to:
 - a. CrP
 - b. Muscle Glycogen
 - c. ATP
 - d. Blood Glucose
6. Overuse of tendons in physical activity can cause problems. Which one of the following problems is associated with tendon overuse?
 - a. Inflammation
 - b. Arthritis
 - c. Hypertrophy
 - d. Bruising
7. After an hour's vigorous exercise program, a 16 year old male, who does not train regularly, experiences some short term effects on his body. Which one of the following would you expect to happen after such a short exercise program?
 - a. increase in cardiac output
 - b. decrease in resting heart rate
 - c. increase in lung volume
 - d. increased breathing rate
8. Which one of the following is an effect of lactic acid during an exercise session?
 - a. Helps you focus on the exercise task
 - b. Causes muscle fatigue during exercise
 - c. Causes stomach ache due to over-eating
 - d. Helps to break down oxygen to produce more energy
9. **SMART** Principle refers to effective goal setting in physical activities. Which one of the following does SMART represent?
 - a. Special; meaningful; achievable; realistic; tedium
 - b. Specific; measurable; achievable; realistic; time-phased
 - c. Sustainable; measurable; agree; recorded; time-phased
 - d. Special; measurable; agreeable; time phased
10. Which one of the following is a long-term effect of exercise on the muscular system?
 - a. An increase in tidal volume in muscles
 - b. A decrease in blood flow to muscle fibres
 - c. An increase in muscle fatigue
 - d. An increase in tolerance to lactic acid

Identify the sentence as True or False:

(Please tick mark the answer)

(5 × 1 = 5)

- | | True | False |
|--|--------------------------|--------------------------|
| 11. Principle of progressive overload states “for gains to occur, one must exercise beyond their existing capacity”. | ☐ | ☐ |
| 12. Fats provide a significant fuel source during high intensity exercise. | ☐ | ☐ |
| 13. Elevation helps reduce injury by helping to support the weight of the limb. | ☐ | ☐ |
| 14. Swimming involves bilateral sequential action of muscle work to produce a powerful and efficient swim stroke. | ☐ | ☐ |
| 15. Task understanding is a key element during cognitive phase of motor learning. | ☐ | ☐ |

Match the following

(5 × 1 = 5)

16.	Circuit training	a. series of exercises of skill activities which are repeated
17.	Skill related fitness variable(s)	b. Exercises that are aerobic and involve dance movements
18.	Aerobic training	c. Agility and Coordination
19.	How many times a week you exercise	d. Long intervals of slow work
20.	The ability to use strength quickly to produce an explosive effort	e. Frequency of exercise
		f. Muscle Endurance and Strength
		g. Power

(16 -), (17-), (18-), (19-), (20-)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

8th Semester of B.P.T. Examination

May 2018

PT403.01 Physiotherapy in Sports & Allied Therapeutics

Date: 07/05/2018

Time: 10:30 AM-01:00 PM

Maximum Marks: 50

Instructions:

1. There are two Sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
2. **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **SECTION – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

SECTION-II

Q. 1. Short answers (5 out of 7) (Answer not exceeding 50 words) (5 × 2 = 10)

- a. Any 2 applications of Newton's Laws of motion in sports
- b. Any 2 advantages of fartlek and circuit training
- c. Any 2 tests for agility and balance
- d. Define oxygen deficit
- e. Define VO₂ Maximum
- f. SAID
- g. Any 2 extrinsic risk factors for sports injury

Q. 2. Short Notes (4 out of 5) (Answer not exceeding 150 words) (4 × 5 = 20)

- a. Principles of training in injury prevention
- b. Energy production for a sprint of 10seconds
- c. Techniques of body composition
- d. Kinematics of throwing
- e. Theories of motor control

Q. 3. Essay (2 out of 3) (2 × 10 = 20)

- a. A 19-year old female swimmer reports an avoidance of carbohydrate in her diet as a means of realizing weight loss. She has achieved only nominal weight loss during the prior 6 months; however, she has noted a continuous decline in her energy and performance, as well as missed and irregular menstruations for 4 months. Discuss in detail about the pathological medical condition affecting her performance and intervention strategies to increase her performance.

- b.** Describe the kinematics of swimming. A swimmer aged 24 reports to the department complaining of pain in anterior shoulder which aggravates during freestyle swim stroke and doing strength training lifting heavy weights and changing swimming gear. Describe the pathomechanical changes in this athlete, risk factors and acute management of the athlete.
- c.** A high jumper comes to you with pain in right anterior knee for the past 3 months. He is able to train regularly but feels intense pain during various phases. You wish to carry out a kinematic study of his technique to understand the nature of problem and the landing patterns. Describe in detail about the phases of jump with emphasis on the pathomechanical changes in this athlete.